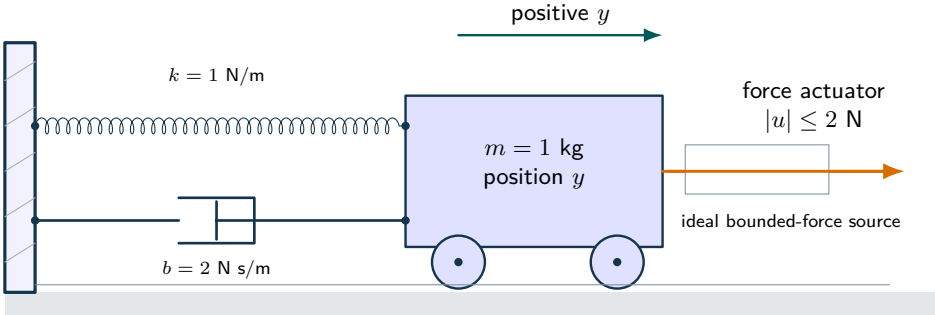

Illustrative force-driven stage; no contact, friction or hard stops

Force and displacement

u: applied force [N].
y: displacement [m].
Requested and applied inputs differ when saturation is active.

Dynamics and bound

$\ddot{y} + 2\dot{y} + y = u$.
$-2 \leq u \leq 2$ N.
The drawn source applies force to the moving mass.

Controller implication

Anti-windup must use applied-minus-requested effort. Wheels depict ideal guidance; rolling friction and hard stops are omitted.